

C Programming Important Questions and Answers

1. Difference between while and do-while loop

while loop	do-while loop
Condition is checked before execution.	Condition is checked after execution.
May execute zero times.	Executes at least one time.
Entry controlled loop.	Exit controlled loop.
No semicolon after condition.	Semicolon is used after condition.

Example:

While Loop Program:

```
#include <stdio.h>

int main()
{
    int i = 1;

    while(i <= 3)
    {
        printf("%d\n", i);
        i++;
    }

    return 0;
}
```

Do-While Loop Program:

```
#include <stdio.h>

int main()
{
    int i = 1;

    do
    {
        printf("%d\n", i);
        i++;
    } while(i <= 3);

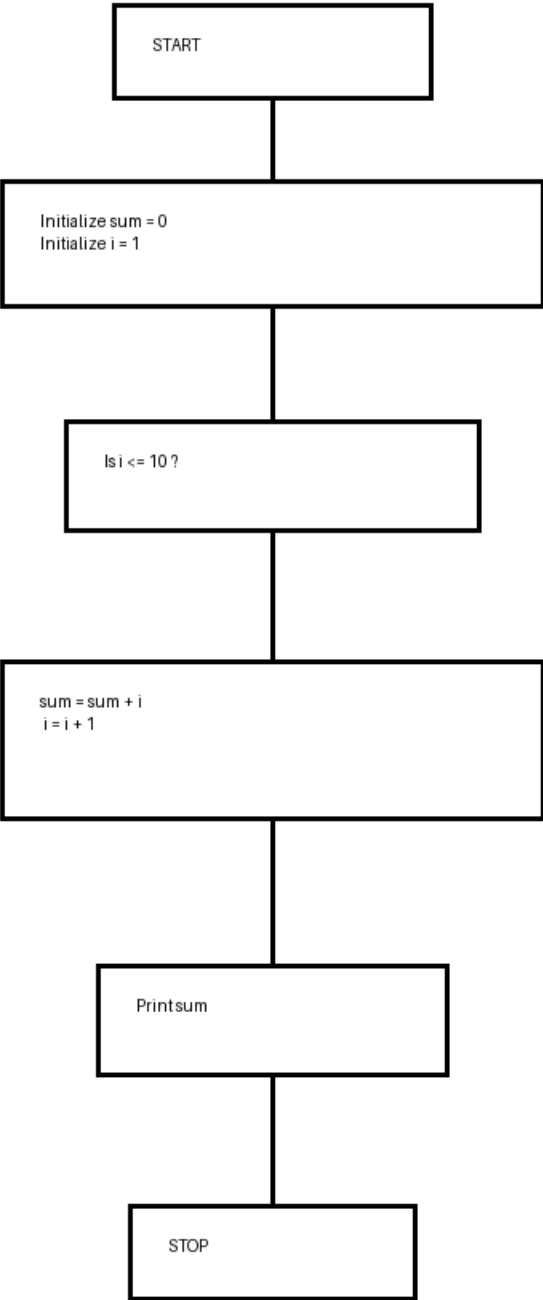
    return 0;
}
```

2. Algorithm and Flowchart for Summation of First 10 Positive Numbers

Algorithm:

1. Start
2. Initialize sum = 0
3. Initialize i = 1
4. Repeat while i <= 10
5. sum = sum + i
6. i = i + 1
7. Print sum
8. Stop

Flowchart:



3. Different Types of Operators in C Language

Operators are symbols used to perform operations on variables and values.

1. Arithmetic Operators

Used for mathematical calculations.

Examples: +, -, *, /, %

2. Relational Operators

Used to compare two values.

Examples: ==, !=, >, <, >=, <=

3. Logical Operators

Used to combine conditions.

Examples: &&, ||, !

4. Assignment Operators

Used to assign values.

Examples: =, +=, -=, *=, /=

5. Increment and Decrement Operators

Examples: ++, --

6. Bitwise Operators

Examples: &, |, ^, ~, <<, >>

7. Conditional Operator

Syntax: condition ? expression1 : expression2

4. C Program to Print First 10 Even Numbers Using for Loop

```
#include <stdio.h>

int main()
{
    int i;

    printf("First 10 Even Numbers:\n");

    for(i = 2; i <= 20; i = i + 2)
    {
        printf("%d\n", i);
    }

    return 0;
}
```

5. C Program to Print First 10 Odd Numbers Using while Loop

```
#include <stdio.h>

int main()
{
    int i = 1, count = 0;

    printf("First 10 Odd Numbers:\n");

    while(count < 10)
    {
        printf("%d\n", i);
        i = i + 2;
        count++;
    }

    return 0;
}
```

6. C Program to Check Whether a Number is Prime or Not

```
#include <stdio.h>

int main()
{
    int n, i, flag = 0;

    printf("Enter a number: ");
    scanf("%d", &n);

    if(n <= 1)
    {
        printf("Not a Prime Number");
    }
    else
    {
        for(i = 2; i <= n/2; i++)
        {
            if(n % i == 0)
            {
                flag = 1;
                break;
            }
        }

        if(flag == 0)
            printf("Prime Number");
        else
            printf("Not a Prime Number");
    }

    return 0;
}
```

7. C Program to Check Whether an Alphabet is Vowel or Not Using switch-case

```
#include <stdio.h>

int main()
{
    char ch;

    printf("Enter an alphabet: ");
    scanf("%c", &ch);

    switch(ch)
    {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
        case 'A':
        case 'E':
        case 'I':
        case 'O':
        case 'U':
            printf("It is a Vowel");
            break;

        default:
            printf("It is not a Vowel");
    }

    return 0;
}
```